Why Flow and Error Control?

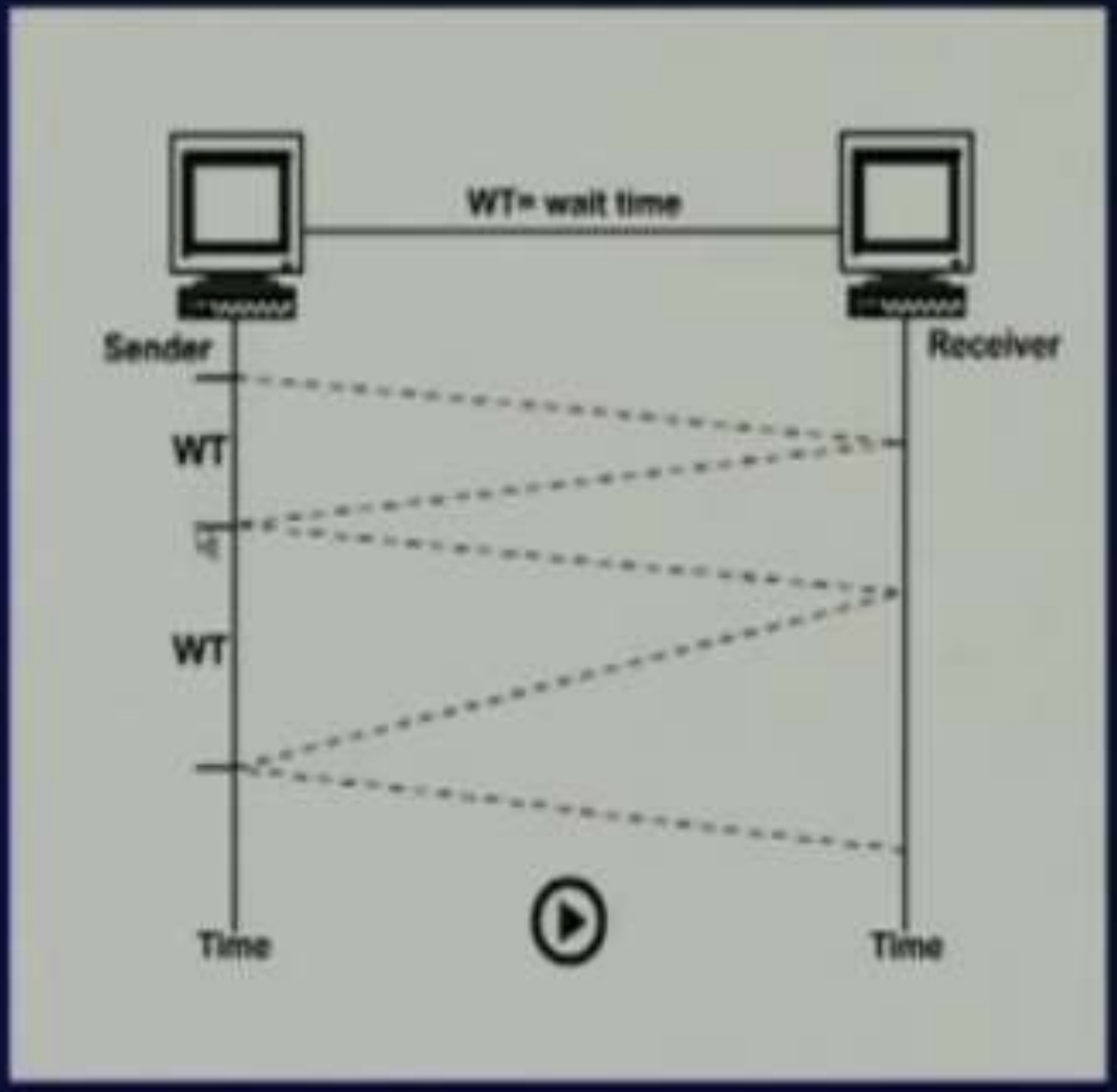
- For reliable and efficient data communication a great deal of coordination is necessary between at least two machines
- Constraints:
  - Both sender and receiver have limited speed
  - Both sender and receiver have limited memory

    Storage
- Requirements:
  - A fast sender should not overwhelm a slow receiver, which must perform a certain amount of processing before passing the data on to the higher level software.
  - If error occur during transmission, it is necessary to devise mechanism to correct it

Stop-and-Wait Flow Control

- The simplest form of flow control.
- ✓ The source transmits a data frame.
- ✓ After receiving the frame, the destination indicates its willingness to accept another frame by sending back an ACK frame acknowledging the frame just received.
- ✓ The source must wait until it receives the ACK frame before sending the next data frame.

Stop-and-Wait Flow Control



Link Utilization in Stop-and-Wait

- *Transmission time*: The time it takes for a station to transmit a frame (normalized to a value of 1).
- *Propagation delay*: The time it takes for a bit to travel from sender to receiver (expressed as *a*).

10^3 bytes

10Mbps

- *a < 1* :The frame is sufficiently long such that the first bits of the frame arrive at the destination before the source has completed transmission of the frame.
- *a > 1*: Sender completes transmission of the entire frame before the leading bits of the frame arrive at the receiver.
- The link utilization U = 1/(1+2a),

  a = Propagation time / transmission time

Sliding Window Flow Control

- With the use of multiple frames for a single message, the stop-and-wait protocol does not perform well.
  - Only one frame at a time can be in transit.
- In stop-and-wait flow control, if $a > 1$, serious inefficiencies result.
- Efficiency can be greatly improved by allowing multiple frames to be in transit at the same time.
- Efficiency can also be improved by making use of the full-duplex line

Sliding Window (sender)

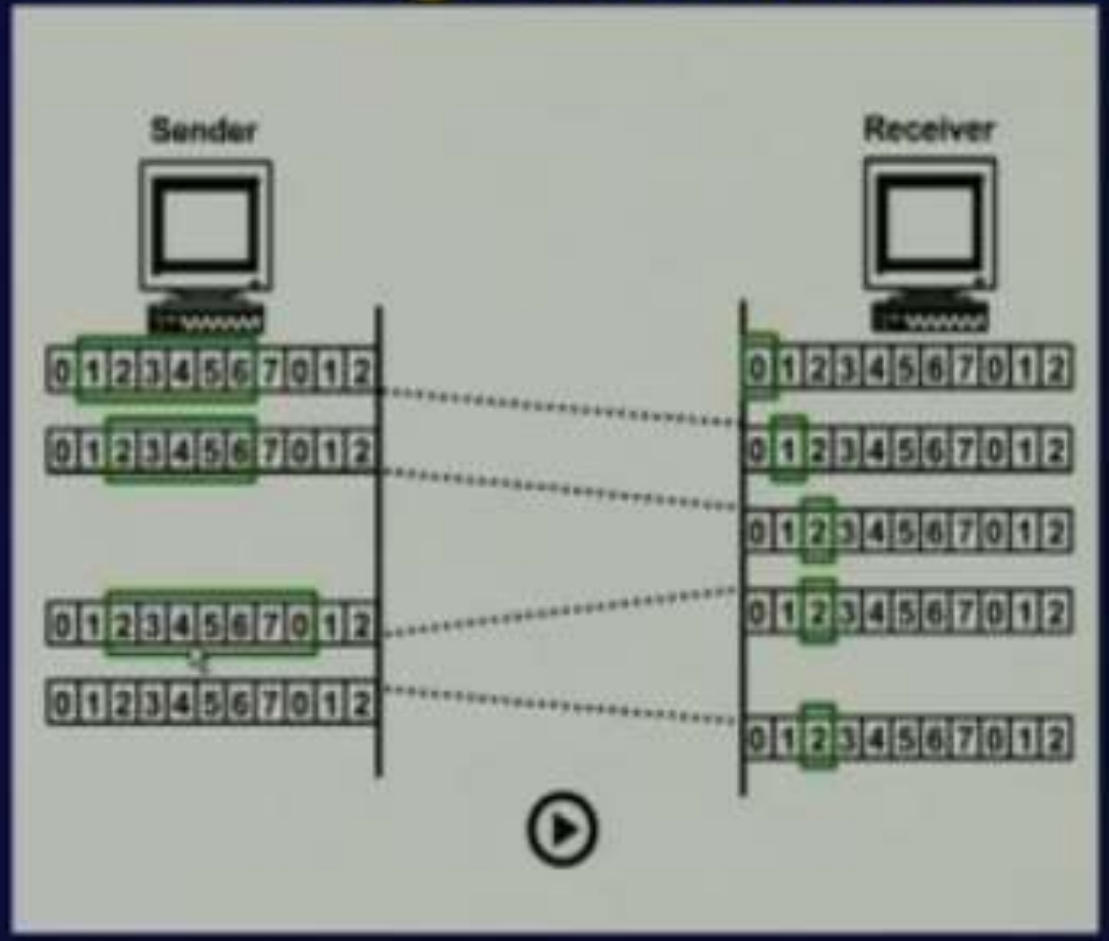
- To keep track of the frames, sender station sends sequentially numbered frames
- Since the sequence number to be used occupies a field in the frame, it should be of limited size.
- If the header of the frame allows k bits, the sequence numbers range from 0 to $2^k - 1$. 2^k
- Sender maintains a list of sequence numbers that it is allowed to send (sender window).
- The size of the senders window is at most $2^k - 1$.
- The sender is provided with a buffer equal to the window size.

Sliding Window (receiver)

- Receiver also maintains window of size 1.
- The receiver acknowledges a frame by sending an ACK frame that includes the sequence number of the next frame expected. This also explicitly announces that it is prepared to receive the next N frames, beginning with the number specified.
- This scheme can be used to acknowledge multiple frames. It could receive frames 2, 3, 4 but withhold ACK until frame 4 has arrived. By returning an ACK5 with sequence number 5, it acknowledges frames 2, 3, 4 at one time.
- The receiver needs a buffer of size 1.

Frames
ACK1
1, 2, 3 ...

Sliding Window Flow Control



$K = 3$

$2^K - 1 = 7$

Piggybacking

$2^k - 1$

- The actual window size need not be the maximum possible size for a given sequence number length.
  - For a 3-bit sequence number, a window size of 4 can be configured.

- If two stations exchange data, each need to maintain two windows. To save communication capacity, a technique called piggybacking is used.
  - Each data frame includes a field that holds the sequence number of that frame plus a field that holds the sequence number used for ACK.
  - If a station has an ACK but no data to send, it sends a separate ACK frame.

Link Utilization in Sliding-window

The link utilization

$$U = \begin{cases} 1, & \text{for } N > 2a + 1 \\ N/(1+2a), & \text{for } N < 2a + 1 \end{cases}$$

Where N = the window size,

and a = Propagation time / transmission time

Example: $K = 5$ $N = 2^5 - 1 = 31$

$a = 10$ $U = 1$ $N > 2a+1$

$a = 200$ $U = 31/1+400$

Backward Error Control

- Model of frame transmission:

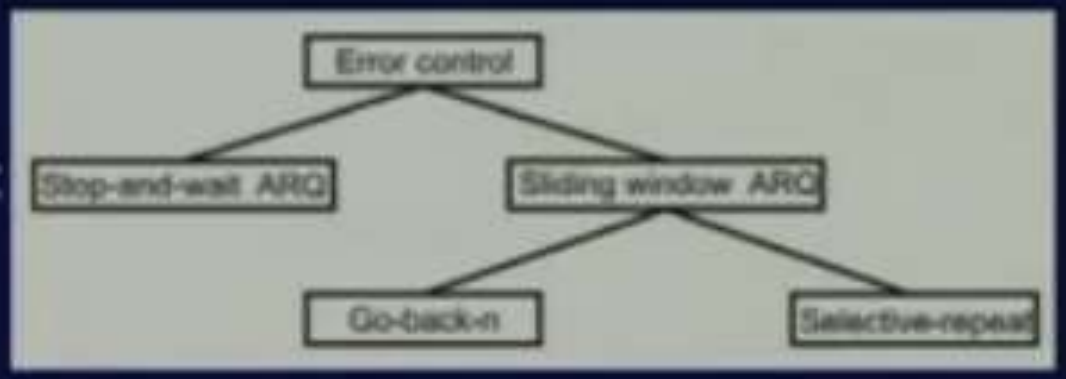
  - Data are sent as a sequence of frames.
  - Frames arrive at the same order as they are sent.
  - Each transmitted frame suffers an arbitrary and variable amount of delay before reception.
- In addition to above, following two types of errors may occur:
- Lost frame: A frame fails to arrive at the other side.
- A noise burst may damage a frame to such an extent that it is not recognizable at the receiving end
- Damaged frame: A recognizable frame does arrive, but some of the bits are in error.

Backward Error Control

- Most common techniques for error control are based on some or all of the following:
  - Error detection: We have already discussed. CRC
  - Positive acknowledgement: The destination returns a positive acknowledgement to successfully received error-free frames. ACK
  - Retransmission after timeout: The source retransmits a frame that has not been acknowledged after a predetermined amount of time.
  - Negative acknowledgement and retransmission: The destination returns a negative acknowledgement to frames in which an error is detected. The source retransmits such frames. NAK

Error Control Techniques

- Collectively, the mechanisms are referred to as Automatic Repeat Request (ARQ).
  - Objective is to turn an unreliable data link into a reliable one.
- Three versions of ARQ are:
  - Stop-and-wait
  - Go-back-N
  - Selective-repeat



Stop-and-Wait ARQ

- Based on the stop-and-wait flow control technique.
- The source station transmits a single frame and then waits for an acknowledgement (ACK).
- No other data frame can be sent until the destination station's reply arrives at the source station.
- To take care of lost and damaged frames, the stations are equipped with:
  - A timer; If no recognizable ACK is received when the timer expires at the end of the time-out interval, then the same frame is sent again.
  - Requires that the transmitter maintain a copy of a transmitted frame until an ACK is received for it.

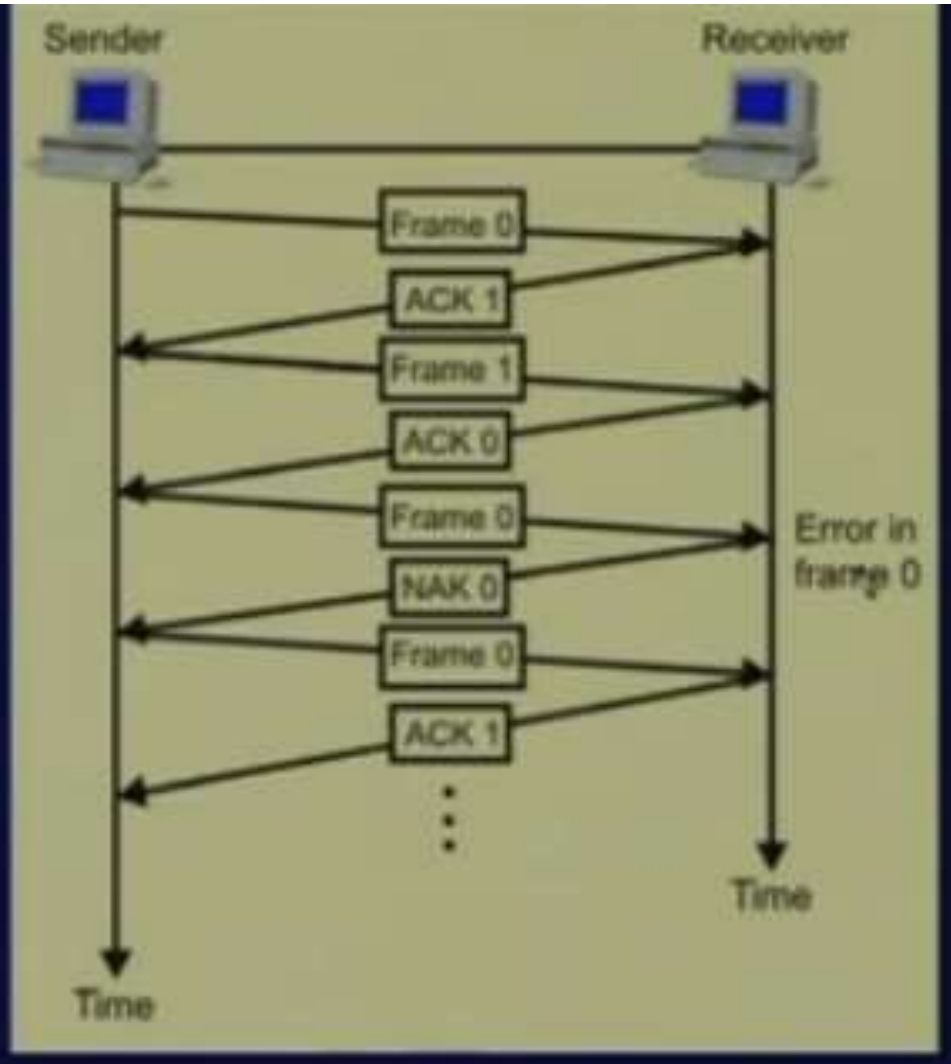
Stop-and-Wait ARQ

- The ACK frame may be damaged.
- The sender will timeout and resend the same frame.
- The receiver receives the same frame twice.
  - How to identify duplicate frames?
- A modulo-2 numbering scheme is used, where frames are alternately labeled with 0 or 1, and positive acknowledgements are of the form ACK0 or ACK1.

Frame 0
Frame 1
Frame 0

Stop-and-Wait ARQ

1-bit



Stop-and-Wait ARQ

- Advantages:

- The main advantage of stop-and-wait ARQ is its simplicity.

- It requires minimum buffer size 1

- Disadvantage: 1

- It makes highly inefficient use of communication links, particularly when 'a' is large

Go-back-N ARQ

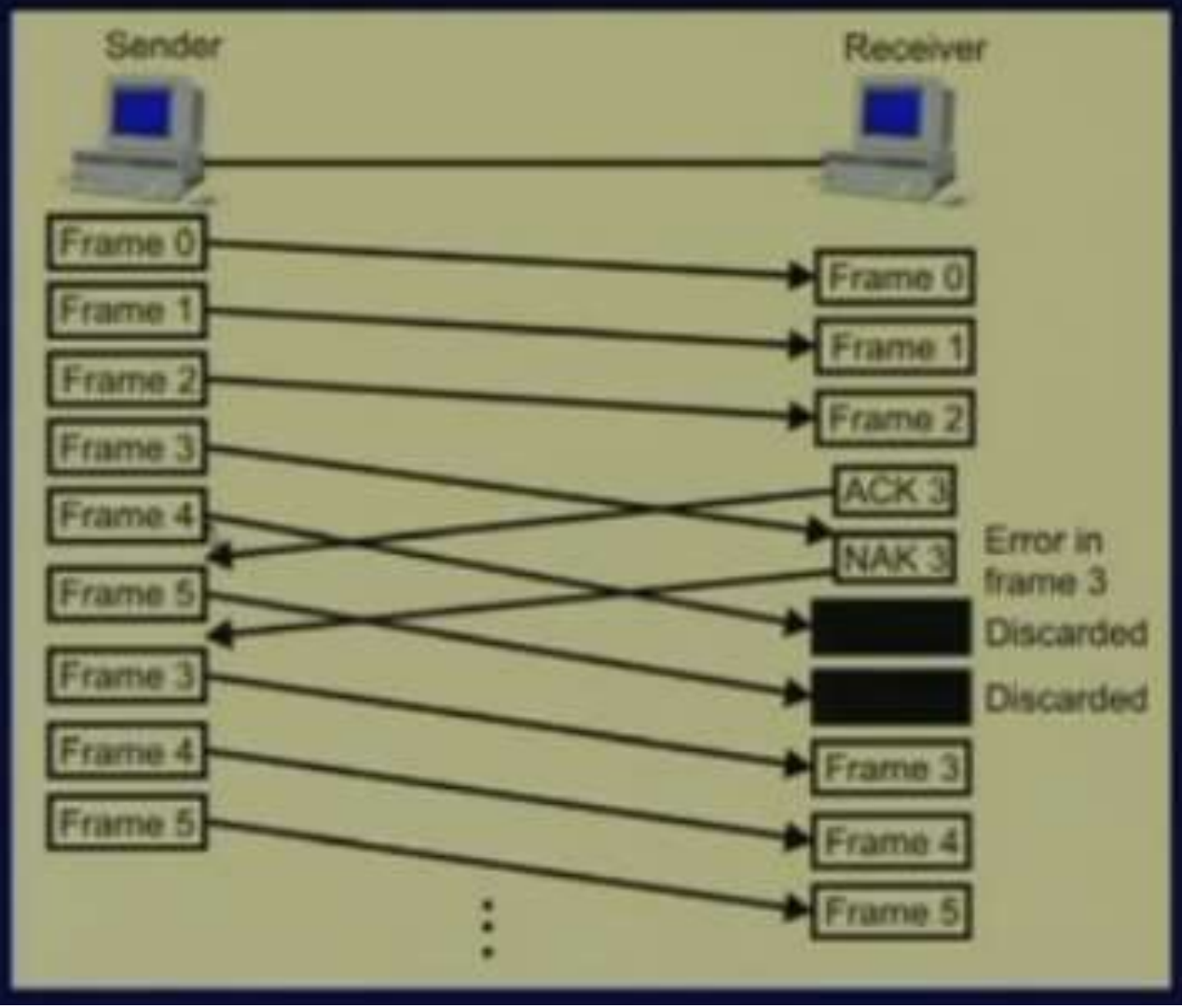
- Based on sliding window protocol. K=3
  - Most commonly used.

- Basic concept:
  - A station may send a series of frames sequentially up to a maximum number
  - The number of unacknowledged frames outstanding is determined by window size, using the sliding-window flow control technique.
  - In case of no error, the destination will acknowledge incoming frames as usual.

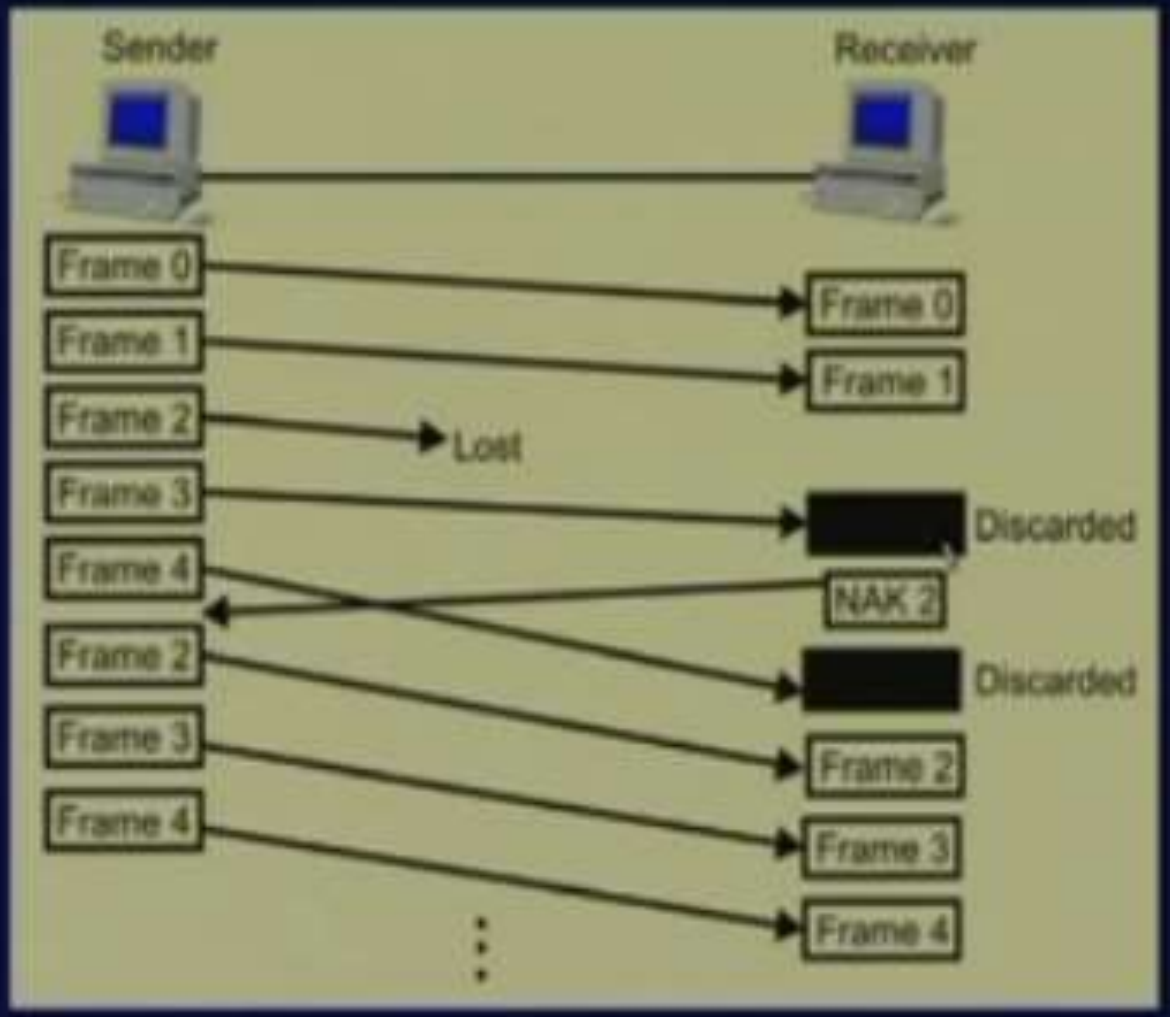
Go-back-N ARQ

- If the destination detects error in a frame, or it receives a frame out of order, it sends a NAK for that frame (using reject or REJ frame).
- The destination will discard the frame in error and all future frames until the frame in error is correctly received.
- The source station, on receiving a REJ, must retransmit the frame in error plus all succeeding frames.

Frame in Error in Go-Back-N ARQ

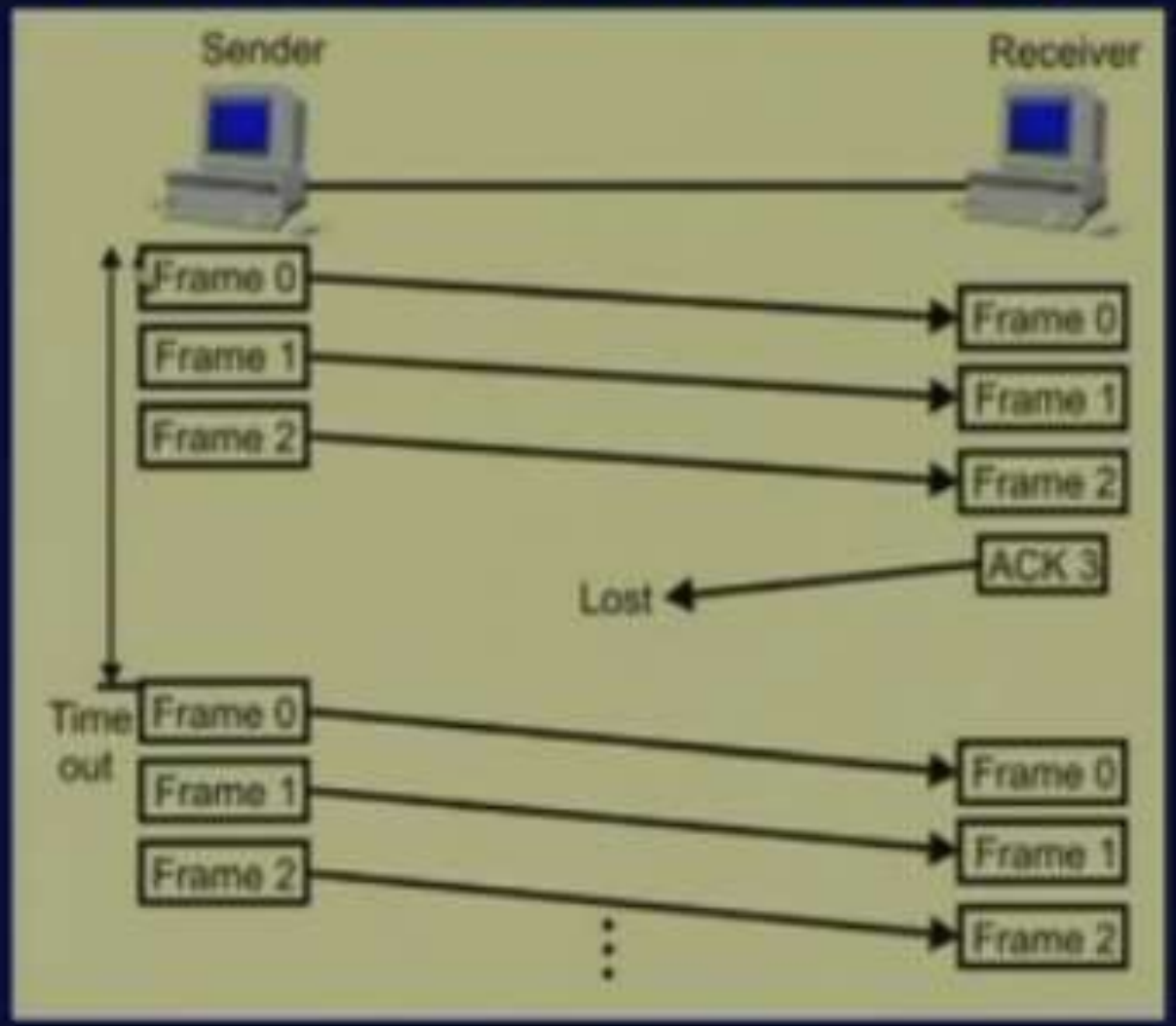


Lost or damaged Frames



2

Lost ACK in Go-Back-N ARQ

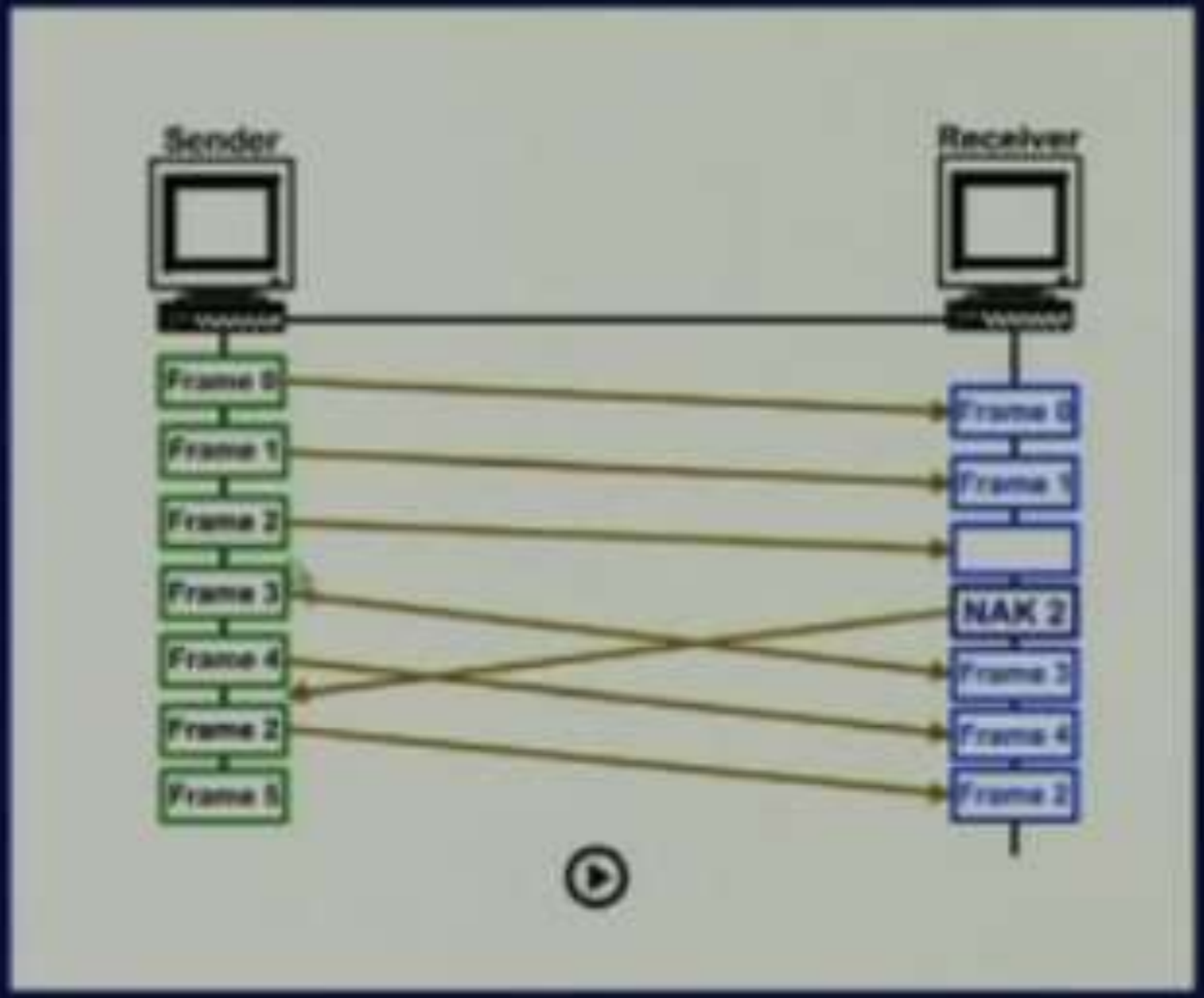


Go-back-N ARQ: window size limit

Assume full-duplex transmission

- The receiving end sends piggybacked acknowledgement by using some number in the ACK field of its data frame. Frames
- Assuming 3-bit sequence number, suppose that a station sends frame 0 and gets back an RR1, and then sends frames 1, 2, 3, 4, 5, 6, 7, 0 and gets another RR1.
- This might either mean that RR1 is a cumulative ACK or all 8 frames were damaged.
  - This ambiguity can be overcome if the maximum window size is limited to 7, i.e. for a k-bit sequence number field it is limited to 2^k-1

Selective Repeat ARQ



Window size limit

- Consider the following scenario with k = 3, and a window size of 7 :
  - Sender sends frames 0 through 6.
  - *Receiver* sends RR7 after receiving all the 7 frames.
  - RR7 gets lost in transit.
  - Sender times out and retransmits frame 0.
  - *The receiver* has already advanced its receive window to receive frames 7, 0, 1, 2, 3, 4, 5.
  - *The receiver* wrongly assumes that frame 7 has been lost and frame 0 is accepted as a new frame.
- The problem can be alleviated by using window size of no more than half the possible sequence number

$K = 3$ $2^3/2 = 4$

- What is data link control?
- Key components of data link control
  - Frame Synchronization
  - Flow Control
  - Error Control
  - Link Management
- High-Level Data Link Control (HDLC)
  - Types of stations
  - Data Transfer modes
  - Frame format

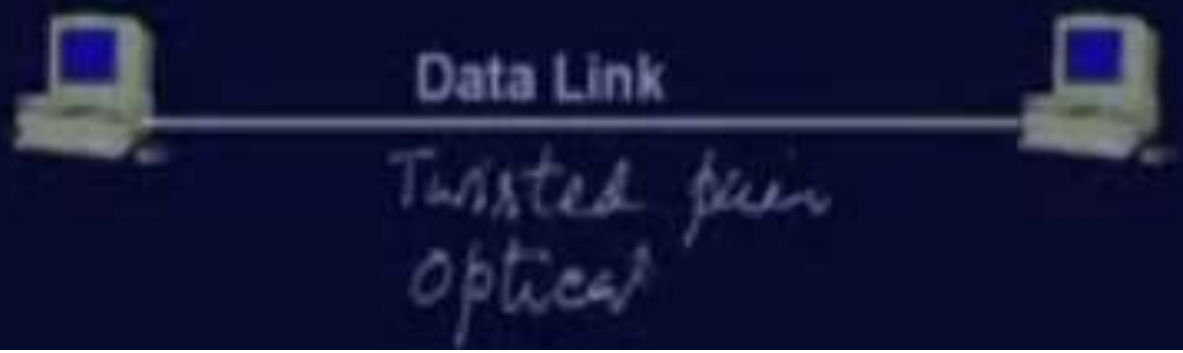
Data Link Control

- Key features of the link:
  - Two machines are directly connected
  - Occasionally errors occur in the communication circuit
  - There is nonzero propagation delay
  - Communication link and the machines have finite data rate
- Objective is to devise suitable mechanism for efficient and reliable communication
- This will require great deal of coordination between the two machines

Data Link Control

–Data Link Control: A layer of logic is added above the physical interface to achieve the necessary control and management.

–Data Link: The transmission medium between systems, when a data link control protocol is used.



Key Components

- **Frame synchronization:** The beginning and end of a data block (called *frame*) should be distinguishable.
- **Flow control:** The sender should not send frames at a faster rate such the the receiver is overwhelmed.
- **Error control:** Any bit errors introduced by the transmission system should be corrected.
- **Control and data on same link:** The receiver must be able to distinguish control information from the data being transmitted.
- **Link management:** Procedures for the management of the initiation, maintenance, and termination of a sustained data exchange.

Frame Synchronization

- When data are transferred from the transmitter to the receiver, unless steps are taken to provide synchronization, the receiver may start interpreting the data erroneously.
- Synchronization levels:
  - Bit, Character and Frame
- Two common approaches:
  - *Asynchronous Transmission*
  - *Synchronous Transmission*

Asynchronous Transmission

- Data are transmitted one character at a time (5-8 bits).
  - Timing or synchronization must only be maintained within each character.
  - The receiver has the opportunity to resynchronize at the beginning of each new character.
- When no character is being transmitted, the line between transmitter and receiver is in an idle state
  - Asynchronous transmission requires 20% or more overhead

Asynchronous Protocols

- Developed for file transfer using asynchronous transmission. Primarily used in modems

File

- Examples: XMODEM, YMODEM, ZMODEM, etc
- Format:
  - One byte start of header (SOH)
  - Two bytes header; 1st byte sequence number, 2nd byte to check the validity of the sequence number
  - 128 fixed data field
  - CRC field for error detection

Synchronous Transmission

- Here, a block of bits or characters is transmitted in a steady stream without start and stop bits.
- The block may be arbitrarily long.
- To prevent timing drift between transmitter and receiver, clock signal is embedded in the data signal (e.g. Manchester encoding) and clock is regenerated at the receiving end
- For sizable blocks of data, synchronous transmission is far more efficient than asynchronous mode.

PLL

20%

Synchronous Protocols

- There is another level of synchronization required, so as to allow the receiver to determine the beginning and end of a block of data.
- Synchronous protocols:
  - Character-oriented BSC
  - Bit-oriented
    - Every block begins with a preamble bit pattern, and generally ends with a postamble bit pattern

Synchronous Frame Format

- A typical synchronous frame format:
  - 8-bit flag (preamble)
  - Control field
  - Data field
  - Control fields
  - 8-bit flag (postamble)
  - The control information, preamble and postamble in synchronous transmission are typically less than 100 bits.

High-Level Data Link Control (HDLC)

- Most important data link protocol.
  - It is the most widely used bit-oriented protocol
  - It is also the basis for many other important data link control protocols.
  - Adopted by ISO committee and also embraced by ITU-T
  - Supports both full-duplex and half-duplex modes in point-to-point and multipoint configurations

HDLC

- HDLC is characterized by:
  - Station Types
  - Configurations
  - Response Modes
  - Frame Formats

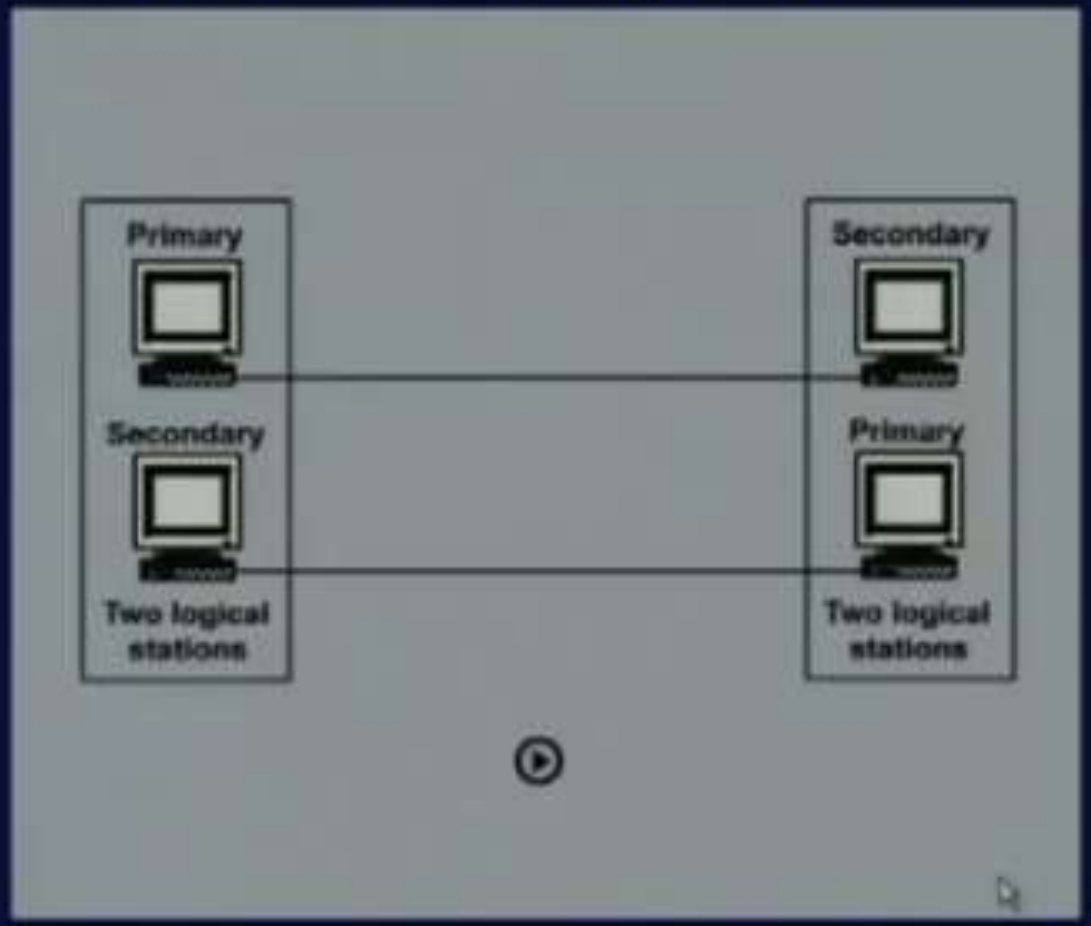
Three types of stations

- Primary Station:
  - Responsible for the operation of the link.
  - Frames issued by primary station are called commands.
- Secondary Station:
  - Operates under the control of a primary station.
  - Frames issued are called responses.
  - Primary maintains a separate logical link with each secondary.
- Combined Station:
  - Combines the features of primary and secondary.
  - May issue both commands and responses.

Link configurations

- Symmetrical Configuration

- Each physical station of a link consists of two logical stations, one a primary and the other a secondary



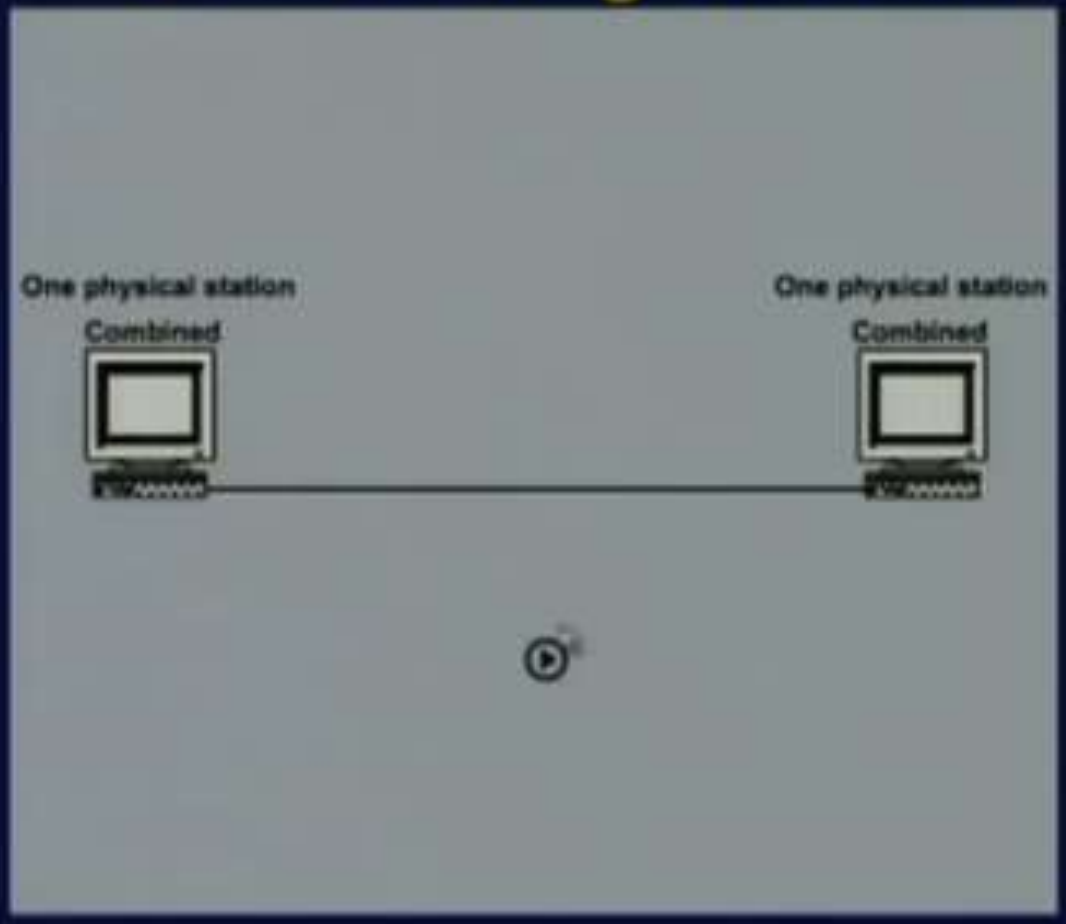
Link configurations

- Symmetrical Configuration

- Each physical station of a link consists of two logical stations, one a primary and the other a secondary

Link configurations

- Balanced Configuration
- Consists of two combined stations.



Response Modes

- Three data transfer modes
  - Normal Response Mode (NRM)
  - Asynchronous Balanced Mode (ABM)
  - Asynchronous response mode (ARM)

Data Transfer Modes

- Normal Response Mode (NRM):
  - Used with unbalanced configuration.
  - Primary may initiate data transfer to a secondary.
  - A secondary may only transmit data in response to a command from primary.
  - Used on multi-drop lines
    - For example, a number of terminals connected to a host computer; the computer polls each terminal for input
  - Also used on point-to-point links
    - if the link connects a peripheral (say, terminal) to the computer

Data Transfer Modes

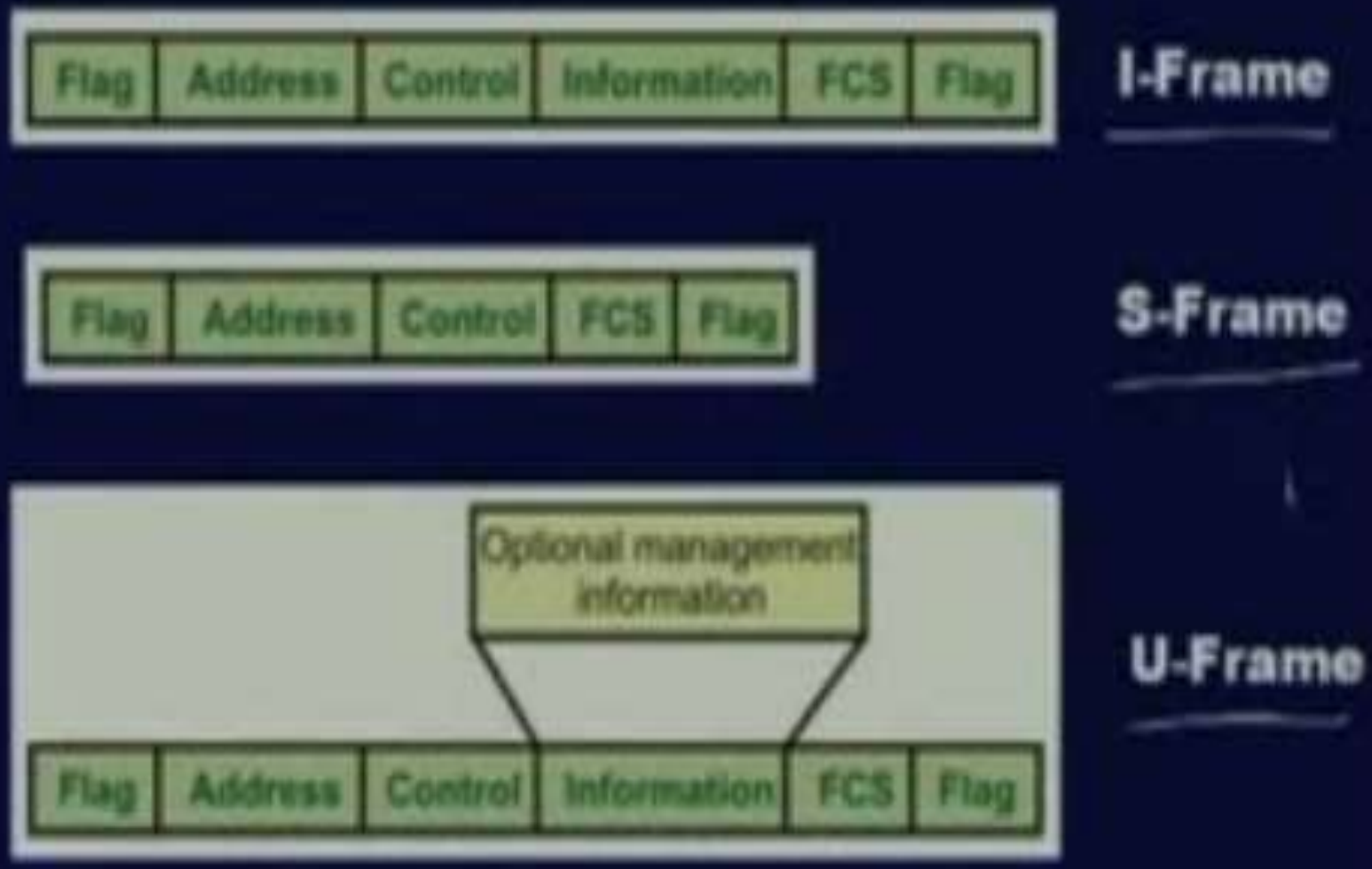
- Asynchronous Balanced Mode (ABM):
- Used with balanced configuration
- Either combined station may initiate transmission.
  - most widely used
  - makes most efficient use of a full-duplex point-to-point link, as there are no polling overheads

Data Transfer Modes

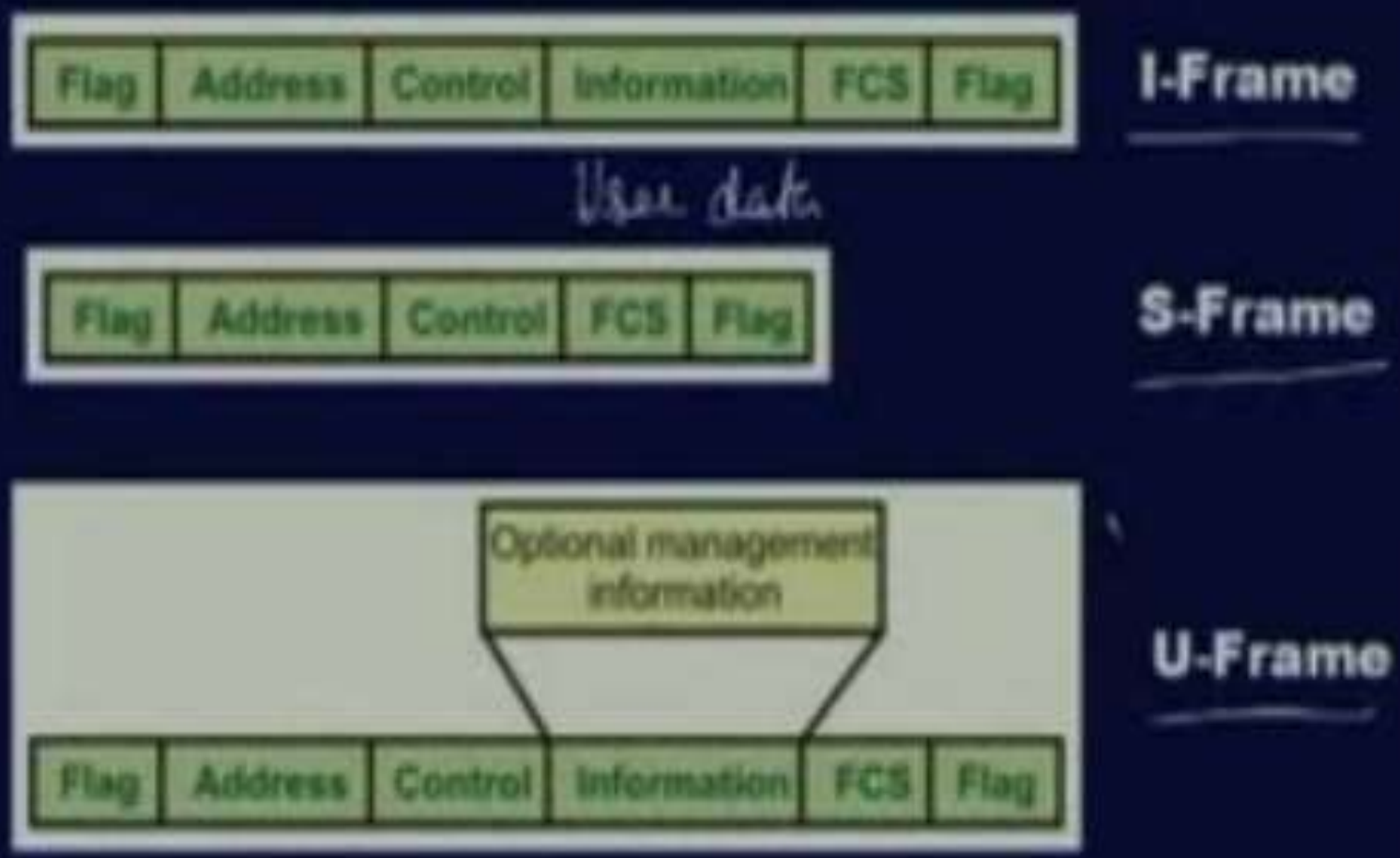
Asynchronous response mode (ARM):

- Used with an unbalanced configuration.
- Secondary may initiate transmission.
- Primary still retains responsibility for the line (initialization, error control, etc)
  - rarely used
  - may be used in very special situations where a secondary may need to initiate transmission

HDLC Frame Types



HDLC Frame Types



Function of Various Fields

Flag	Address	Control	Information	FCS	Flag

- Flag Fields
  - Each frame starts and ends with the special bit pattern 01111110
  - Bit stuffing is used to unambiguously identify the flag fields
  - When the same flag is used to mark both the beginning and the end,
    - A 1-bit error may merge two frames into one
    - A 1-bit error inside the frame could split it into two.

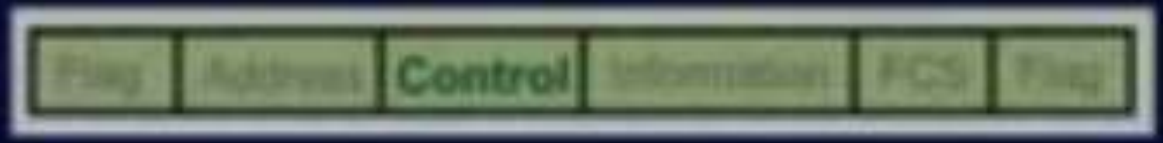
Function of Various Fields

Flag	Address	Control	Information	FCS	Flag

- Address Field
  - identifies the secondary station that transmitted the frame (from address) or is to receive the frame (to address)
  - this field is not needed for point-to-point links, but is included for the sake of uniformity
  - both single-octet and multi-octet addressing are possible.

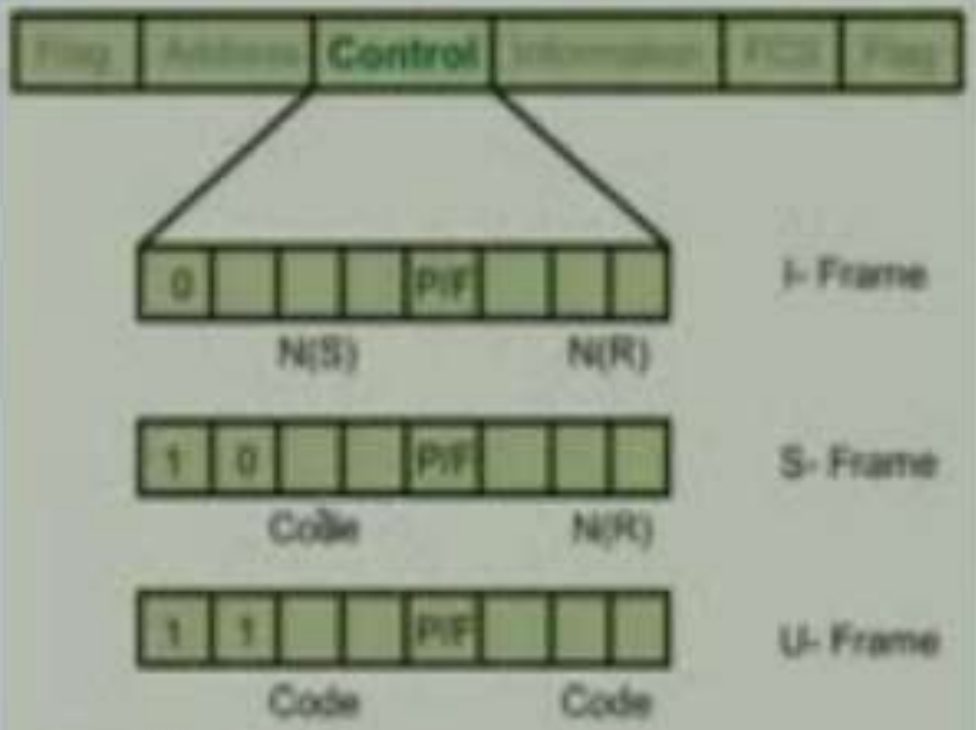
128

Function of Various Fields



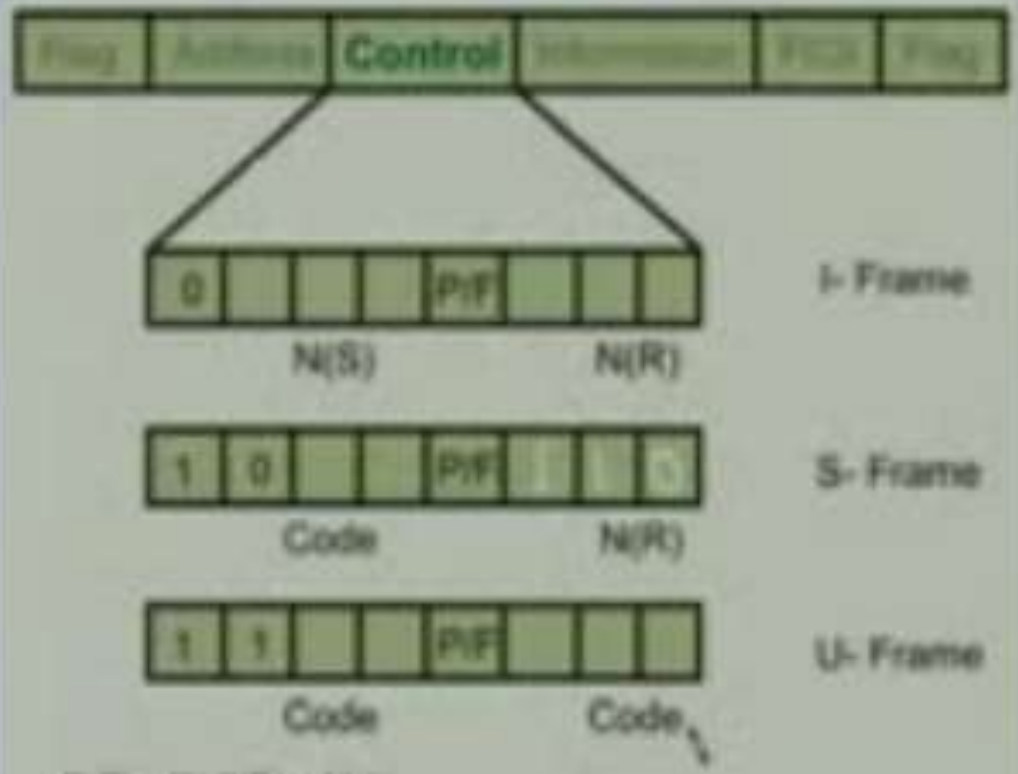
- **Control Field:** Defines three types of frames
  - Information frames (I-frames) carry the data to be transmitted; flow- and error-control information are also piggybacked using ARQ mechanism
  - Supervisory frames (S_frames) provide the ARQ mechanism when piggybacking is not used
  - Unnumbered frames (U-frames) provide supplemental link control functions
  - I- and S-frames use 3-bit sequence numbers.

Function of the Control Field



P/F = Poll/Final bit
N(S) = Sequence number of frame sent
N(R) = Sequence number of frame expected
Code = For S and U frames

Function of the Control Field



P/F = Poll/Final bit
N(S) = Sequence number of frame sent
N(R) = Sequence number of frame expected
Code = For S and U frames

1 0 1

Code ACK

0 0 RR

0 1 Rej

1 0 RNR

1 1 SREJ

control & Management

Function of Various Fields

Flag	Address	Control	Information	FCS	Flag

- **Information Field:** Present in only I-frames and some U-frames

User data

  - Can contain any sequence of bits (multiple of 8), but is usually limited in length from considerations of error control
  - Frames with empty I-field are transmitted continuously on idle point-to-point lines in order to keep the connection alive.

Function of Various Fields

Flag	Address	Control	Information	FCS	Flag

- **Information Field:** Present in only I-frames and some U-frames
  - Can contain any sequence of bits (multiple of 8), but is usually limited in length from considerations of error control
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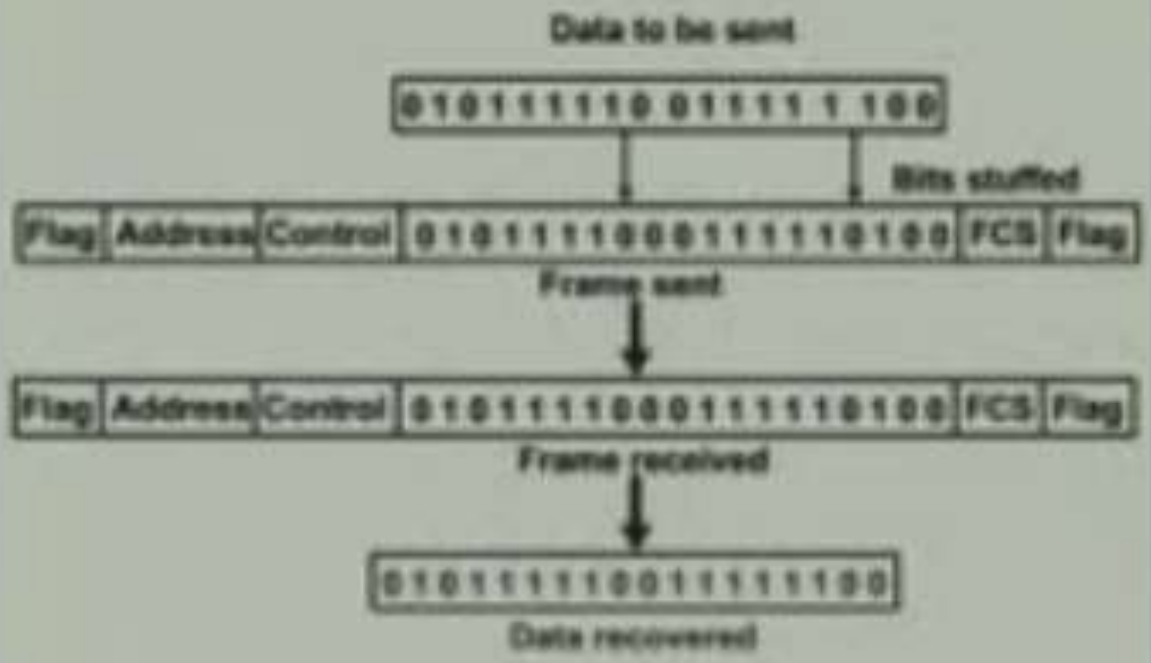
Bit Stuffing



Function of Various Fields

- Frame Check Sequence Field: 16- or 32-bit CRC code computed using address, control & data fields.

Link Access Procedures

- Several protocols have been developed which are subset of HDLC
  - Examples:
  - Link Access Procedure, Balanced (LAPB): It is used in Integrated Services Digital Network (ISDN) B channel using Asynchronous Balanced Mode (ABM) to connect two devices of combined type.
  - Link Access Procedure for D channel (LAPD): Used in ISDN using ABM.
  - Link Access Procedure for Modems (LAPB): Developed to provide HDLC features to modems and designed to do synchronous-asynchronous conversion, error detection and retransmission

Answer to the Questions of LEC-17

3. In HDLC, what is bit stuffing and why is it needed?

Ans: The presence of the bit-sequence 01111110, which is used as flag to indicate the start and end of a frame, may lead to division of a single frame in to more than one frames wrongly. This problem is overcome by using bit-stuffing. A 0 is introduced after after each occurrence of five consecutive 1's in the information field of a frame.

4. What is piggybacking? How is it used in HDLC?

Ans: In case of full-duplex communication, the receiver of a frame may send ACK as part of the information frame sent to the other side. This is known as piggy-backing. In HDLC, the control field of a I-frame has got two subfields; N(S) and N(R), each of 3 bits. The N(S) is used to convey the sequence number of the frame transmitted and N(R) is used to provide the acknowledgement number, i.e. the sequence number of the frame the receiver is expecting.